

Standard Model Particles — BST Substrate Annotations

27 particles + Standard Model masses from D_{IV^5} substrate

Quarks (6)

Gen 1

u
2.16MeV
rank-2 winding

d
4.67MeV
rank-2 winding

Gen 2

c
1.27GeV
speaking pair 1

s
93MeV
speaking pair 1

Gen 3

t
173GeV
speaking pair 2 + Yukawa

b
4.18GeV
speaking pair 2

Leptons (6)

e
0.511MeV
 $N_c = 3$ generations

ν_e
< eV
seesaw = 17

μ
105.66MeV
 $T190 : (24/\pi^2)^6$

ν_μ
< eV
seesaw = 17

τ
1776.86MeV
 $T2003 : 49 \cdot 71$

ν_τ
< eV
seesaw = 17

Gauge + Higgs (5)

γ
massless
 $T2478: U(1)_{em}$

g
massless
 $SO(5) \rightarrow SU(3)$

$W^\pm$
80.4GeV
EW SSB

Z^0
91.2GeV
EW SSB

H
125.1GeV
 $T2478: SSB$ origin

Quark color

$SO(5)$ sub-group $\rightarrow SU(3)$ color

Lepton mass ratios

$T190 + T2003 + (24/\pi^2)^6$

Gauge groups

$K = SO(5) \times SO(2) \rightarrow SM$

EW SSB origin

$T2478 + SO(2)$ preserves vev

No SUSY / monopoles / GUT

Five Absences (Casey-named)

Composite: $m_p/m_e = 6\pi^5 \approx 1836.118$ (0.002% D-tier, T190)

BST primary integers underlying Standard Model:

$\text{rank} = 2$ (gen pairing) • $N_c = 3$ (color + generations) • $n_C = 5$ (orbitals) • $C_2 = 6$ (Yang-Mills) • $g = 7$ (f orbitals) • $N_{\text{max}} = 137$ (α^{-1})

a_e CROWN JEWEL ppt precision (Vol 2 Ch 8)